RB2301 – Robot Programming – Final Project

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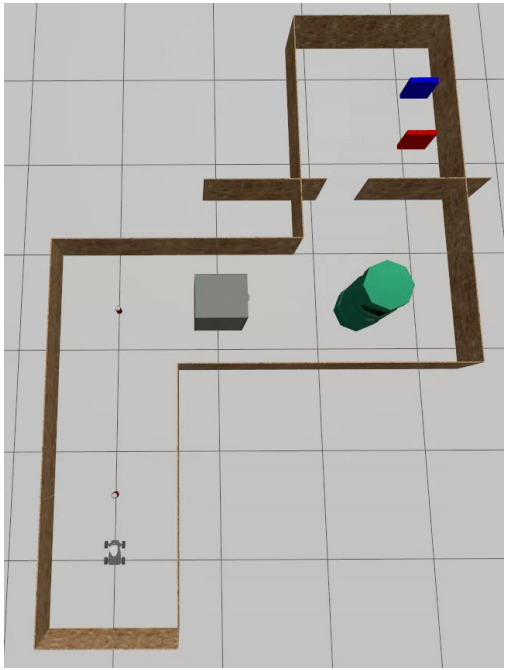


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“Real-World” Obstacle Course

- CA1: reactive control (short-term thinking).
- CA2: path planning (deeper reasoning).

- Final Project:
  - General “concept” of the map, but **no precise measurements**
    - Aim: robust/resilient approach
  - LiDAR available, and external tracking (position/heading)
  - Three sections:
    - Zig-Zag Corridor with Static Obstacles
    - Small Door
    - Dynamic Swinging Obstacles
  - Any approach allowed :-D



Final Project Map

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FP – Simulation Map

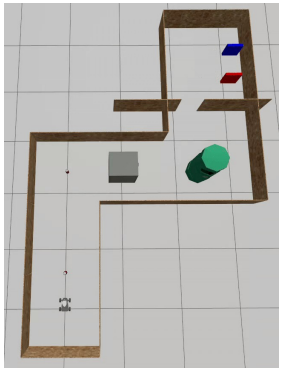
- Set randomisation variables in ***obstacle_modifier.py***

```
randomise_gate_position = True
randomise_zigzag_size = True
randomise_obstacle_speed = True

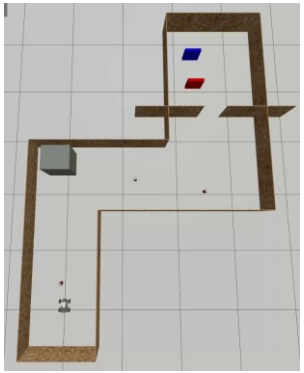
spawn_static_obstacles = True
randomise_static_obstacles = True
```

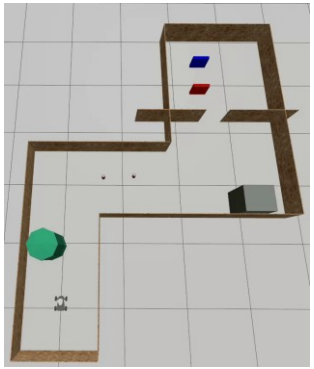
  - Original map in *obstacle_course_world_fp_original.sdf*
  - DO NOT modify the original map

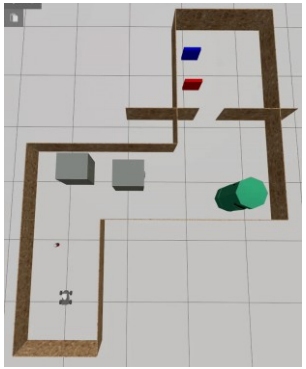
- Note: Moving obstacles do not have collision
  - Adding collision is too janky in Gazebo, I tried :c



Original Map







Map Variations

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FP – Sense, Think, Act

- Sense
  - Pose data – From either Odom (Gazebo) or Optitrack (real-life)
  - LiDAR data – Remember to start rplidar when running on real-life hardware
- Think
  - No Occupancy grid given for Final Project
  - Walls are guaranteed rectilinear
- Act
  - *move_2D()*
  - Extra marks for heading (?)

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Let's Set Everything Up Together!

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FP – Getting the Code

- Download rb2301_fp zip file from canvas
- Transfer the downloaded files into your existing RB2301 workspace
  - *fp.sh*
  - *gz_fp.sh*
- Transfer the downloaded package folders into your RB2301 workspace's *src/* folder
  - *rb2301_fp/*
  - *rb2301_gz/*
  - Feel free to copy and replace your ca1 *rb2301_gz* folder; the new *rb2301_gz* folder contains data and launch files for all CAs and Final Project
- Change directory to your RB2301 workspace and build the workspace
  - E.g. *cd ~/Documents/rb2301*
  - *colcon build* ***--symlink-install***

Add --symlink-install to:

1. Have linked files work properly
2. Not need to rebuild after editing source

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FP – Setup Development Environment

- Open the *rb2301_fp* folder in your preferred IDE
- Open and edit *obstacle_course.py*
- Add your logic to the *timer_callback()* function of *ObstacleCourseNode*

```
def timer_callback(self):
    """Controller loop. Insert path planning and
    if self.pose is None:
        print("No pose detected")
        return # Does not run if no pose received

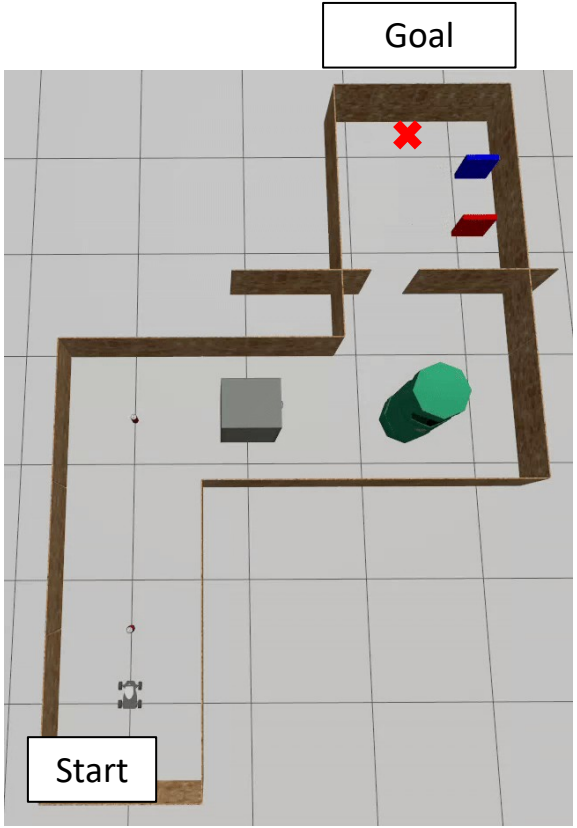
    ###### INSERT CODE HERE ######
    self.get_logger().info(f"Pose: {self.pose}")
    self.move_2D(0.1)
    ###### INSERT CODE HERE ######
```

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FP – Gazebo Sim

- In a terminal, navigate to your RB2301 workspace
  - E.g. *cd ~/Documents/rb2301*
- Run the Gazebo simulation for Final Project
  - *./gz_fp.sh*
- Run your path planning code on another terminal
  - *./fp.sh*
- Goal is to reach past the moving obstacles
- If you wish to launch Gazebo with different starting coordinates, modify the x and y values in *./gz_fp.sh*

```
ros2 launch rb2301_gz fp_gazebo.launch.py x:=0.0 y:=0.0
```

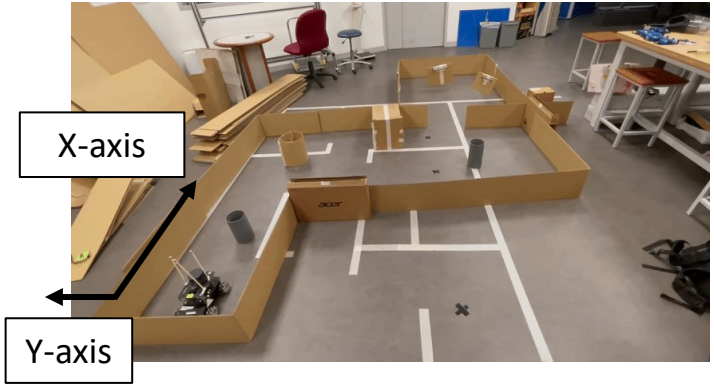


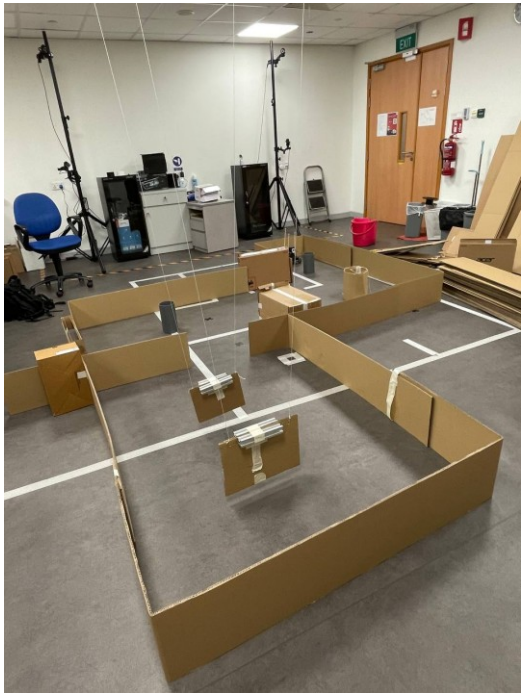
Gazebo map

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FP – Real-life test

- Swinging obstacles will be released when your robot is at the gate
- Will run 2x for each group
  - 1x with the same maze configuration
  - 1x with slightly randomised positions and dimensions for each group





For yours (and everyone's) safety, please do **not** step inside the maze, and be especially careful around the swinging obstacles.

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FP – Code Updates

- No longer need to set *is_simulation* manually
  - Script automatically detects whether it's in Gazebo or on hardware (using LiDAR scan size)
  - You can still use *self.is_simulation* to have different behaviours for Sim vs Real, if you need

```
def main(args=None):
    print("Starting obstacle course")
    rclpy.init(args=args)

    # Use lidar scan length to determine if in simulation or not
    lidar = LidarNode()
    while lidar.is_simulation is None:
        rclpy.spin_once(lidar)
    is_simulation = lidar.is_simulation
    lidar.destroy_node()
```

- Still need to set optitrack topic to `/vrpn_mocap/bingda_00x/pose` where x is the number of your robot

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CA2 – Hardware Updates

- Connect to local network on your laptop
  - WiFi name: **TP-RMI-Lab**, Password: **rmi_pass**

- SSH to robot and create workspace
  - *ssh bingda@192.168.1.20x* → Replace x with the # of your Sparklife robot (E.g. x=2 for BINGDA-002)
  - Password for robot: **bingda**
  - *mkdir -p ~/Downloads/rb2301_fp_<Your Name>/src* □ Differentiate your packages using your name

- rsync *src* folder to robot (From your laptop's **native terminal**)
  - *rsync -auvx --delete <source> <external name>@<external ip>:<external path>*
  - Example: *rsync -auvx --delete ~/Documents/rb2301_ca2/src/rb2301_fp bingda@192.168.1.20x:~/Downloads/rb2301_fp_<Your Name>/src*
  - Copy Shell script: *rsync -auvx --delete ~/Documents/rb2301_ca2/fp.sh bingda@192.168.1.20x:~/Downloads/rb2301_ca2_<Your Name>*

Copy the shell script over

- Build your workspace on robot (On **ssh'ed terminal**)
  - *cd ~/Downloads/rb2301_ca2_<Your Name>*
  - *colb* → Added alias for *colcon build --symlink-install*

Alias for colcon build

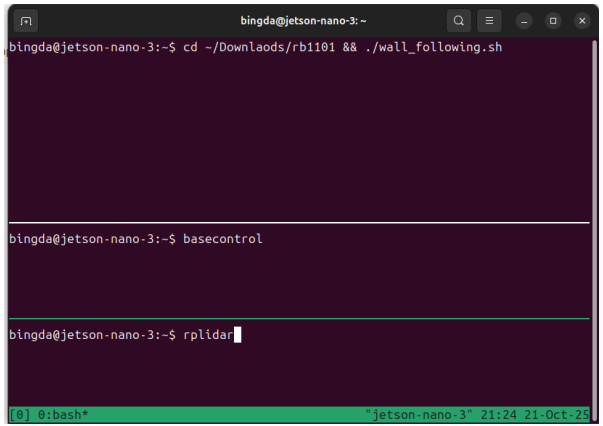
- Run each of these in a separate terminal (On **ssh'ed terminal**)
  - *basecontrol* → Added alias for *ros2 launch base_control_ros2 base_control.launch.py*
  - *rplidar* → Added alias for *ros2 launch rplidar_ros rplidar.launch.py*
  - *vrpn* → Launches the vrpn node using alias for *ros2 launch vrpn_mocap client.launch.yaml server:=192.168.1.199 port:=3883*
  - *cd ~/Downloads/rb2301_fp_<Your Name> && ./fp.sh* → Change directory to your workspace and run the shell script

Alias for common commands

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FP - Updates

- tmux now installed on all robots
- In SSH'ed terminal, enter *tmux* → tmux allows you to run multiple terminals from just 1 SSH
  - No longer need to SSH 3 times to run basecontrol, rplidar and your code
- Once in tmux, hit **CTRL + SHIFT + B + "** keys to split the terminal



Terminal with 3 tmux splits

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That’s all folks, have fun!

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